A Literature-Implied Upgrade for Quaternion Compressed Sensing

Run `fa_banach_001` – lane 1

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Status. Literature-implied answer. This note records a direct identification with a later block-sparse recovery theorem; it is not claimed as a new mathematical result.

Source question

Badeńska and Błaszczyk prove stable quaternion ℓ_1 recovery for a real sensing matrix under $\delta_{2s} < 1/3$ [1, Theorem 4.1 and Corollary 5.1]. Immediately afterward, on PDF page 12, they write that they conjecture this requirement is not optimal.

Supporting theorem

Gao and Ma prove that mixed ℓ_2/ℓ_1 minimization stably recovers block sparse signals when the block restricted isometry constant satisfies

$$\delta_{2s|\mathcal{I}} < \frac{4}{\sqrt{41}} \approx 0.6246,$$

and in the noiseless, exactly block-s-sparse case the recovery is exact [2, Theorem 1 and Corollary 1, PDF p. 3].

Identification

Write each quaternion coordinate as

$$z_j = a_j + b_j\mathbf{i} + c_j\mathbf{j} + d_j\mathbf{k}$$

and let $R : \mathbb{H}^n \to \mathbb{R}^{4n}$ collect (a_j, b_j, c_j, d_j) as the jth four-dimensional block. Then

$$\|z\|_{\ell_1(\mathbb{H}^n)} = \sum_{j=1}^{n} |z_j| = \sum_{j=1}^{n} \|Rz[j]\|_2 = \|Rz\|_{2,1}.$$

If $\Phi \in \mathbb{R}^{m\times n}$, realification of $z \mapsto \Phi z$ is, up to a coordinate permutation,

$$A = \Phi \otimes I_4 : \mathbb{R}^{4n} \longrightarrow \mathbb{R}^{4m}.$$

It preserves Euclidean data and noise norms. Hence the quaternion program in [1] is precisely Gao–Ma's mixed ℓ_2/ℓ_1 program for the block partition $\mathcal{I} = (4, \ldots, 4)$.

It remains only to compare the two restricted isometry constants. If u is supported on at most k blocks and $u^{(0)}, \ldots, u^{(3)} \in \mathbb{R}^n$ are its four component vectors, then each $u^{(r)}$ is k-sparse and

$$\|Au\|_2^2 = \sum_{r=0}^{3} \|\Phi u^{(r)}\|_2^2, \qquad \|u\|_2^2 = \sum_{r=0}^{3} \|u^{(r)}\|_2^2.$$

Summing the scalar RIP inequalities shows $\delta_{k|\mathcal{I}}(A) \leq \delta_k(\Phi)$. Conversely, embed any real k-sparse vector in the first component and set the other three components to zero. This gives the reverse inequality, and therefore

$$\delta_{k|\mathcal{I}}(\Phi \otimes I_4) = \delta_k(\Phi).$$

Applying Gao–Ma with $k = 2s$ proves stable quaternion recovery, and exact recovery in the noiseless sparse case, whenever

$$\boxed{\delta_{2s}(\Phi) < 4/\sqrt{41}}.$$

Since $4/\sqrt{41} > 1/3$, this fully confirms the source's stated nonoptimality conjecture in its real-measurement setting.

Provenance and scope

This is an agent-identified implication. Gao and Ma do not mention quaternion signals or the source paper. The threshold $4/\sqrt{41}$ is recorded only as a certified strict improvement, not as the optimal threshold. Quaternion-valued sensing matrices are outside the scope.

A bounded search used the exact source title, "quaternion compressed sensing RIP", "block sparse recovery", and the δ_{2s} threshold terminology. It found Gao–Ma's theorem and later block-RIP refinements, but no paper explicitly advertising this quaternion specialization.

References

[1] A. Badeńska and Ł. Błaszczyk, *Compressed sensing for real measurements of quaternion signals*, arXiv:1605.07985; J. Franklin Inst. 354 (2017), 5753–5769.

[2] Y. Gao and M. Ma, *A new bound on the block restricted isometry constant in compressed sensing*, J. Inequal. Appl. 2017, article 174, doi:10.1186/s13660-017-1448-2.